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Subtype-MMCC: multimodal contrastive clustering approach for cancer subtype discovery with multi-omics data

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Abstract

The diversity and complexity of cancer pose significant challenges in creating target treatment strategies. Identifying molecular subtypes of cancer is crucial for recognizing patients with distinct molecular profiles, thereby enhancing the accuracy of diagnosis, prognosis, and treatment decisions. With recent advancements in technology, there's been a significant increase in the availability of multi-omics data, which is instrumental in the understanding of different cancer subtypes. However, accurately subtyping cancer is difficult due to the high dimensionality and heterogeneity of omics data. Current research in subtype identification often consolidates multi-omics data into a single dataset through simple concatenation and then employs machine learning models to derive a lower-dimensional representation, neglecting the unique distributions of different omics data types. Additionally, they separate representation learning and clustering into two stages, initially learning latent representations and then applying clustering algorithms, leading to suboptimal results due to overlooking the intrinsic clustering structures in the initial learning phase. To address these limitations, we propose a novel deep unsupervised learning model, Subtype-MMCC (Multi-modal Contrastive Clustering) that combines a multi-modal architecture with decoupled contrastive clustering to create an end-to-end framework. Tested on eight TCGA cancer datasets, Subtype-MMCC outperforms existing clustering methods, with its efficacy further validated by survival and clinical analysis outcomes.

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1. Introduction

Cancer is a complex and diverse genomic disease that serves as a leading cause of death worldwide, accounting for nearly 10 million deaths in 2020 [1]. Due to high heterogeneity, cancer patients exhibiting identical phenotypic traits frequently possess varied genomic modifications, significantly affecting their clinical results and future prognosis [2].

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Therefore, accurate identification of cancer subtype is vital for cancer diagnosis and prognosis, as well as to enhance clinical treatment and precision medicine, ultimately leading to improved patient survival.

The rapid advancement in high-throughput sequencing and experimental biotechnologies has led to a significant accumulation of omics (refers to comprehensive sets of molecular biology information like genetic sequences and protein expressions used to study biological systems) and clinical data, offering a deeper insight into the complex mechanisms of various cancers [3, 4]. Projects like The Cancer Genome Atlas (TCGA)[5] and International Cancer Genome Consortium (ICGC) [6] have been instrumental in this development. These initiatives provide a vast array of omics data, including genomic, epigenomic, transcriptomic, and proteomic information, from thousands of samples across various cancer types, thus offering significant opportunities to study the occurrence and development of cancer. Despite the substantial increase in the availability of omics data and clinical data, there remains a scarcity of definitive ground-truth labels for cancer subtypes, thereby making cancer subtype identification, a challenging research problem.

Early studies in cancer subtyping predominantly involved applying standard clustering algorithms to single-omic data to determine cancer subtypes [7]. Individual omics data, which characterizes molecular features at specific levels, can restrict the accuracy of results. Therefore, the integration of multi-omics data is essential, as it offers a more comprehensive perspective on the complex interactions among biomolecules across various levels [3]. This will lead to a deeper understanding of the pathogenic mechanisms of cancers and enhance cancer subtyping accuracy.

The integration, analysis, and interpretation of omics data have emerged as a compelling area in cancer research. Broadly, these methodologies fall into three primary categories: early integration, late integration, and intermediate integration [3]. Early integration, the simplest approach, concatenates omic matrices to form a single matrix with features from multiple omics, and applies a single-omic clustering algorithm to the concatenated matrix. K-means, LRAcluster, and Spectral clustering techniques belongs to this category [3]. The notable limitation of these methods is that they ignore the distribution difference and dimensional difference between omics data, and increase the dimension of input data and exacerbate the dimension problem. In Late integration, each omic dataset is clustered separately and the different clustering's are finally integrated to obtain a single solution. COCA [8] and PINS [9] belong to this category. The limitation of this is that, relying solely on clustering solutions in the integration phase, can result in the loss of weaker signals that are present individually in each omic dataset.

Intermediate integration methods, represent a more sophisticated approach that combines dimensionality reduction and clustering into a single, comprehensive model. Unlike basic concatenation or simple merging of results, these methods effectively reduce the complexity of high-dimensional data while preserving essential information [3, 10]. Intermediate integration can be further divided in to three categories: statistical methods, similarity-based methods and deep learning methods [3, 10]. Statistical methods model the probabilistic distribution of samples, often grouping them into distinct clusters based on data distribution. These methods can incorporate biological knowledge into their models, using techniques like Bayesian priors or specific probabilistic functions for different types of data. Eg: NMF [11], MCCA [12], iCluster [13], iCluster+ [14] and iClusterBayes [15]. However, the heterogeneity of multi-omics data, with its varying types and distributions, poses a challenge for accurate modeling in statistical methods, which also risk oversimplifying complex interactions and missing subtle, biologically important patterns. Similarity-based methods like Similarity Network Fusion (SNF) [16] and NEMO [17], involve constructing individual similarity networks for each omics dataset and then integrating these into a unified network. This approach offers a comprehensive view of sample relationships across various data types, facilitating effective cancer subtyping and clustering analysis. However, these methods can be sensitive to data noise and network parameters, as they rely on the stability of kernel distance functions, which can lead to variability in results with minor changes in data or parameters.

Recently, Deep learning has demonstrated promising outcomes in healthcare research, owing to its advanced learning capabilities, adaptability, and proficiency in handling complex medical datasets [18]. Subtype-GAN [4] employs a deep adversarial learning approach to effectively characterize multi-omics data with varying distributions. This model uses consensus clustering combined with a Gaussian mixture model on the latent representations it learns for identifying cancer subtypes. Subtype-Former [19] employs Multilayer Perceptrons (MLPs) and Transformer blocks to compress high-dimensional multi-omics data and then utilizes K-means to predict the accurate clusters. Limitation of studies [4, 19] is that the representation learning step is independent of the subsequent clustering step, and the representation from the first step is not guaranteed to be suitable for the second step. Subtype-DCC [10] address this limitation by introducing an end-to-end clustering technique that combines deep learning and decoupled contrastive learning [20], ensuring that both representation learning and clustering are jointly optimized for more effective and

suitable subtype identification in high-dimensional multi-omics data. Another limitation of all studies [4, 19, 10], is that they start the modelling by integrating multiple high-dimensional, heterogeneous omics datasets through simple concatenation, effectively treating it as a single dataset. This approach leads to a latent low-dimensional representation that fails to account for the unique distributions inherent in different types of omics data. By overlooking the distinct characteristics and distributional variances of each omics dataset, these studies may not fully capture the complexity and nuances essential for accurate analysis and interpretation in high-dimensional biological data contexts.

We propose a novel method Subtype-MMCC: utilizing multi-modal architecture that treats each omics dataset as a separate modality and is integrated with decouple contrastive learning approach used in Subtype-DCC to create an end-to-end deep learning modal. Subtype-MMCC adeptly maintains the unique characteristics of each individual omics dataset while simultaneously capturing the collective essence of all datasets, ensuring both specificity and comprehensiveness in data analysis.

2. Methodology

2.1. Datasets & Data pre-processing

The dataset for our study was obtained from The Cancer Genome Atlas (TCGA) [5]. We analyzed eight cancer datasets, namely Breast (BRCA), Head and Neck (HNSC), Kidney (KIRC), Lung Adenocarcinoma (LUAD), Lung Squamous (LUSC), Prostate (PRAD), Skin (SKCM) and Stomach (STAD). By following prior studies [4, 19, 10] we decided to utilized multi-omics data spanning four molecular platforms: Copy Number (Examines gene copy number variations in the genome, indicating genetic changes linked to cancer), DNA methylation (Studies DNA methylation changes affecting gene expression, related to cancer types and subtypes), miRNA (Focuses on small non-coding RNA molecules regulating gene expression, crucial for cancer biology and subtype differentiation) and mRNA (Analyzes RNA expression patterns to identify gene expression changes across different cancer subtypes). The raw data for these four omics types underwent pre-processing, including feature selection and normalization as per previous work [19]. This process yielded 3105 copy number, 3217 DNA methylation, 383 miRNA, and 3139 RNA values for subsequent analysis. Additionally, we also collected clinical data of respective samples for the purpose of evaluating subtyping performance.

2.2. The Subtype-MMCC Model

Multimodal Encoder Architecture : In our proposed model for integrating multi-omics data, we deviate from previous methods [4, 19, 10] that merely concatenated multiple high dimensional, heterogeneous datasets. Instead, we introduce a multi-modal encoder architecture designed to capture and preserve the unique characteristics of each omics dataset. By processing each dataset Copy Number, DNA methylation, miRNA and miRNA through a dedicated encoder, we generate high-level feature representations. These features are further refined by Multi-Layer Perceptron (MLPs) projectors, resulting in distinct modalities for each dataset (fig 1 - Network architecture with multiple modalities). Additionally, we combine the outputs of all projectors to form a unified feature vector (denoted by Fig 1 - Sample level MLP), enabling our model to analyze shared characteristics across datasets effectively. This approach ensures that both dataset-specific traits and inter-dataset shared features are meticulously captured and utilized for enhanced analytical precision. The encoder architecture in our model employs Multilayer Perceptrons (MLPs) comprising layers with 5000, 2000, 1000, and 256 neurons, respectively, using ReLU activation functions to introduce non-linearity. For projection, we utilize two-layer MLPs, also incorporating ReLU activations to facilitate non-linear processing of features for enhanced learning efficiency.

Decoupled Contrastive learning on Multimodal Architecture : In recent years, Contrastive learning which facilitates models to identify distinct data features effectively, has demonstrated state-of-arts performance in the field of unsupervised representation learning [21]. It works by refining models to recognize and emphasize the distinctions and commonalities within data, thus improving data representation. This method strategically adjusts data embeddings to cluster similar pairs closer and disparate ones further apart, fostering more precise pattern recognition in

datasets without labeled information. Constative learning has been effectively applied for clustering with omics data in previous work [10]. In this work, we are adopting decouple constarstive clustering on a multi-modal architecture.

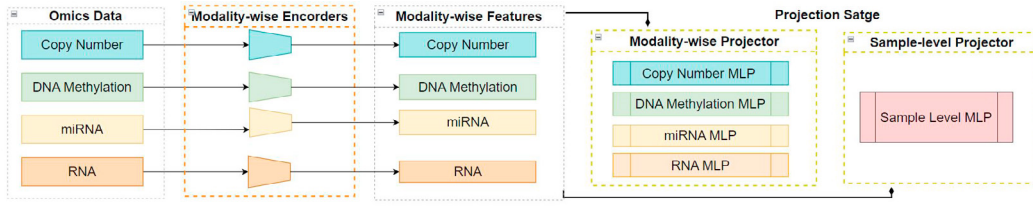


Fig. 1. Network architecture of Subtype-MMCC

First, we do pair-construction backbone independently for each omics dataset. Each data instance undergoes specific augmentations—Noise, Mask, and Dropout—to generate two correlated yet varied samples. Noise adds standard Gaussian noise to simulate measurement errors; Mask obscures parts of the matrix at random, mimicking missing data scenarios; and Dropout removes matrix elements at a specified rate to prevent overfitting by reducing dependency on any single input feature. A unified deep neural network, denoted as $f(\cdot)$, is employed to extract features from augmented data samples, resulting in two sets of correlated features, $h_i^a = f(x_i^a)$ and $h_i^b = f(x_i^b)$. This process ensures that features derived from different augmentations of the same data instance are captured, facilitating the model's learning from the inherent variability within the data.

Instance-level contrastive head (ICH) : This module independently assess the similarity within augmented positive pairs and minimize it across negative pairs [21]. This process, applied to each modality, enhances the accuracy of instance-level representations. Sample pairs are constructed utilizing pseudo-labels derived from data augmentations. When a data sample, represented as x_i , undergoes augmentation, it results in two correlated versions: x_{ai} and x_{bi} . These two variants of the same original sample then constitute a positive pair, denoted as $\{x_{ai}, x_{bi}\}$. Conversely, negative pairs are generated by pairing augmented samples that originate from different original samples, thus establishing a basis for the model to differentiate between similar (positive) and dissimilar (negative) data points. Given a mini-batch of size N , data augmentation is performed on each sample x_i , resulting in $2N$ data samples: $\{xa_1, \dots, xa_N, xb_1, \dots, xb_N\}$. For each specific sample xa_i , there are $2N - 1$ total pairs, among which the pair formed by xa_i and its augmented counterpart xb_i ($\{xa_i, xb_i\}$) is considered a positive pair. This leaves $2N - 2$ pairs as negative, involving xa_i and all other samples except its own augmented pair.

In computing the loss, we used decoupled contrastive loss to traditional cross-entropy loss (InfoNCE loss) due to its effectiveness in mitigating the negative-positive-coupling effect, which can reduce learning efficiency, especially in scenarios with smaller batch sizes [10]. By removing the positive pairs from the loss function's denominator (Equation 1 [21]), decoupled contrastive loss (Equation 2, 3 [10]) enhances learning efficiency and improves performance with less sensitivity to batch size, thereby reducing the computational resources required for model training. In Subtype-MMCC, the Instance-Level Contrastive Head (ICH) is individually modeled for each modality.

$$l_a^i = -\log \frac{\exp(s(z_i^a, z_i^b)/\tau)}{\sum_{j=1}^N [\exp(s(z_i^a, z_j^a)/\tau) + \exp(s(z_i^a, z_j^b)/\tau)]} \quad (1)$$

$$l_{DCI}^a = -\log \frac{\exp(s(z_i^a, z_i^b)/\tau)}{\sum_{j=1}^N [\exp(s(z_i^a, z_j^a)/\tau) + \exp(s(z_i^a, z_j^b)/\tau)]} \quad (2)$$

$$l_{DCI}^a = -s(z_i^a, z_i^b)/\tau + \log \sum_{j=1, j \neq i}^N [\exp(s(z_i^a, z_j^a)/\tau) + \exp(s(z_i^a, z_j^b)/\tau)] \quad (3)$$

Cluster-level contrastive head (CCH) : The Cluster-Level Contrastive Head is designed to enhance cluster distinction within the embedding space. It employs soft labels that represent the probability of each sample's membership

to clusters, with the goal of maximizing intra-cluster similarities and minimizing inter-cluster similarities. Mathematically, this is achieved using cosine similarity measures between soft labels, formulated as $s(\hat{y}_i^a, \hat{y}_i^b)$ to compare the affinity of a sample to a cluster across augmented views.

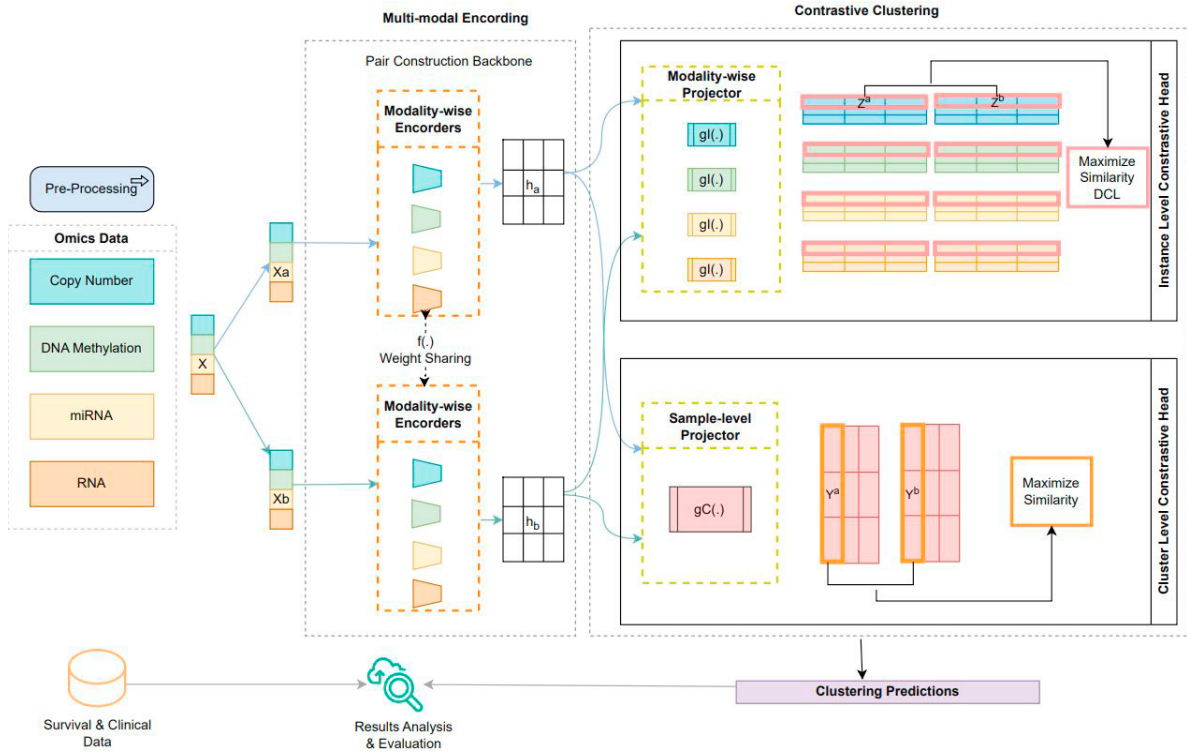


Fig. 2. The framework of Subtype-MMCC

The loss function for each cluster C_i is computed as Equation 4 [21]), using a softmax function weighted by a temperature parameter τ_c across all clusters except for the one in question. This is summed up to derive the overall cluster-level loss as shown in Equation 5 [21] (where $H(Y)$ and $P(\hat{y}_k)$ are given by Equation 6 and 7 respectively), which also includes an entropy term to enforce confident and distinct cluster assignments.

$$\hat{l}_i^a = -\log \left(\frac{\exp(s(\hat{y}_i^a, \hat{y}_i^b)/\tau_c)}{\sum_{j=1}^M \exp(s(\hat{y}_i^a, \hat{y}_j^a)/\tau_c) + \exp(s(\hat{y}_i^a, \hat{y}_j^b)/\tau_c)} \right) \quad (4)$$

$$L_{clu} = \frac{1}{2M} \sum_{i=1}^M (\hat{l}_i^a + \hat{l}_i^b) - H(Y) \quad (5)$$

$$H(Y) = \sum_{i=1}^M [P(\hat{y}_i^a) \log P(\hat{y}_i^a) + P(\hat{y}_i^b) \log P(\hat{y}_i^b)] \quad (6) \quad P(\hat{y}_i^k) = \frac{\sum_{t=1}^N y_{ti}^k}{\|Y^k\|_1}, \quad k \in \{a, b\} \quad (7)$$

The cluster-level contrastive head is formulated using the combined projector (sample level MLP), ensuring that the calculation of the contrastive loss leverages the integrated representations to enhance cluster differentiation. By optimizing this loss function, the Subtype-MMCC model ensures a clear separation of clusters, which is fundamental for accurate subtype identification in complex datasets. The output layer of the combined projector is tailored to correspond to the number of subtypes present in tumor samples. The number of clusters (subtypes) of is set to a fair value obtained from previous studies for each tumor type [22, 23, 24]. Consequently, cluster labels are determined by the Cluster-level Contrastive Head (CCH) upon completion of the analysis. The Subtype-MMCC framework is

demonstrated in Fig 2 consisting of Encoder and projector based Multimodal architecture, Decoupled contrastive clustering unit consisting of instance-level and cluster-level contrastive heads.

Loss Analysis and Interpretability : The Instance-Level Contrastive Head (ICH) component's loss is derived by aggregating the losses across all four individual modalities. This integration is key to capturing the unique characteristics of each data type while maintaining coherence in the learning process. Both the ICH and Cluster-Level Contrastive Head (CCH) components are jointly learning and optimizing together in an end-to-end fashion during training. This concurrent optimization ensures that learning is both integrated and cohesive, aligning the objectives of representation learning and clustering more effectively than if they were optimized separately. The overall loss for the Subtype-MMCC model is calculated using the objective function L (Equation 8), which incorporates a series of weights (denoted as w_1 to w_6) to each component. These weights play a crucial role in balancing the contributions of different components and modalities within the model, allowing for a tailored approach to learning that can adapt to the specific nuances of the multi-omics data being analyzed.

$$L = w_5(w_1 L_{CN-ins} + w_2 L_{meth-ins} + w_3 L_{miRNA-ins} + w_4 L_{RNA-ins}) + w_6 L_{clu} \quad (8)$$

In our experimental methodology, we rigorously determined the optimal values for weights w_1 to w_6 . This involved plotting individual loss curves for each modality within the Instance-Level Contrastive Head (ICH) and for the combined modalities in the Cluster-Level Contrastive Head (CCH). By analyzing these curves, we observed the progression of loss reduction during training. Additionally, we employed t-SNE visualizations for each omic dataset based modality, enhancing our understanding of how each omics dataset contributes to the optimization of the model. Crucially, these t-SNE plots also enhance the interpretability of our model, a key aspect of modern deep learning frameworks. They provide a visual representation of how different omics datasets influence cluster formation, offering insights into the underlying structures learned by the model and making the complex workings of deep learning more accessible and comprehensible.

2.3. Evaluation metrics & Comparison Algorithms

To evaluate the clustering performance, we decided to use a combination of evaluation metrics utilized by previous works [3, 4, 19, 10]. First, we used Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI) which respectively evaluate the consistency between the obtained cluster labels and ground-truth labels and the similarity between the true clustering and the predicted clustering. A larger value of them indicates a better clustering result. Then we used two important evaluation criteria's proposed by previous study [3]: the survival difference between different subtypes and the enrichment of the number of significant clinical parameters. The differential survival between clusters was quantified using $-\log_{10}$ P-values, obtained through the log-rank test [25]. This metric operates under the assumption that patient subtypes hold biological significance when there's a marked difference in their survival rates. The enrichment of clinical labels in the clusters is done by testing the number of significant clinical parameters. Six clinical labels were selected for this enrichment analysis: age at diagnosis, gender, pathologic stage, pathologic T, N, and M classifications. To ensure accuracy in our survival analysis, especially against the impact of small sample sizes, we employed permutation tests for calculating the P-values in the log-rank test. We also conducted enrichment analyses on clinical parameters using the chi-square test and the Kruskal-Wallis test [3]. Additionally we also plotted survival analysis curves and t-sne graphs to visualize the clustering results.

For comparison, we choose 10 of the latest cancer subtyping algorithms, including early, late, and immediate integration methods and statistical and similarity based methods as well as state-of-art deep learning methods. Namely: LRACluster[26], MCCA [12], moCluster[27], NEMO [17], CIMLR [28], iClusterPlus [13], PINS [9], Subtype-GAN [4], Subtype-Former [19] and Subtype-DCC [10].

3. Results and Discussion

3.1. Comparison with existing methods

Analyzing the performance of multi-omics clustering methods, Subtype-MMCC has demonstrated commendable results. In metric NMI and ARI (Table 1), it achieves the highest NMI scores in two datasets but also stands as the

Table 1. Results NMI & ARI : Highlighted Optimal (**Bold**) and Second Optimal (Underlined)

Modal	BRCA	HNSC	KIRC	LUAD	LUSA	PRAD	SKCM	STAD
CIMLR	0.300/0.211	0.226/0.200	0.103/0.102	0.090/0.051	0.079/0.048	0.033/-0.012	0.037/-0.022	0.039/0.016
i-cluster	0.006/-0.000	0.003/-0.006	0.007/-0.001	0.019/0.002	0.032/0.006	0.022/-0.003	0.006/-0.003	0.008/0.001
LRA-cluster	0.367/0.254	0.149/0.100	0.081/0.085	0.117/—0.077	0.321/0.276	0.142/0.139	0.030/0.029	0.228/0.238
MCCA	0.361/0.226	0.291/0.229	0.079/0.074	0.229/0.164	0.202/0.130	0.462/0.511	0.026/-0.009	0.168/0.102
mocluster	0.352/0.221	0.192/0.136	0.069/0.053	0.203/0.104	0.214/0.187	0.205/0.114	0.016/0.003	0.069/0.017
NEMO	0.459/0.291	0.285/0.252	0.166/0.151	0.329/0.239	0.417/0.451	0.332/0.323	0.069/0.040	0.370/0.364
PINS	0.344/0.224	0.294/0.245	0.209/0.222	0.000/0.000	0.000/0.000	0.188/0.153	0.000/0.000	0.061/0.075
Subtype-GAN	0.401/0.254	0.299/0.279	<u>0.144/0.135</u>	0.332/0.231	0.190/0.153	0.231/0.131	0.043/0.037	0.353/0.198
Subtype-Former	<u>0.481/0.326</u>	0.419/0.408	0.137/0.134	<u>0.349/0.287</u>	0.321/0.346	0.036/-0.003	<u>0.049/0.040</u>	<u>0.353/0.362</u>
Subtype-DCC	0.476/0.294	0.371/0.375	0.126/0.104	0.374/0.293	<u>0.249/0.265</u>	0.028/-0.001	0.052/0.035	0.223/0.141
Subtype-MMCC	0.497/0.327	<u>0.403/0.378</u>	0.155/0.147	0.298/0.231	0.229/0.243	<u>0.081/0.030</u>	0.045/ 0.058	0.279/0.219

second-highest in two others, indicating its adeptness at aligning with true data classifications. The performance of Subtype-MMCC in the HNSC and KIRC datasets is particularly noteworthy, although it's overall results on NMI metric is slightly below Subtype-Former. But in ARI metric, Subtype-MMCC's capabilities shine through by-passing other ten methods. Subtype-MMCC has the best results in three datasets and followed closely as the second-best in another three.

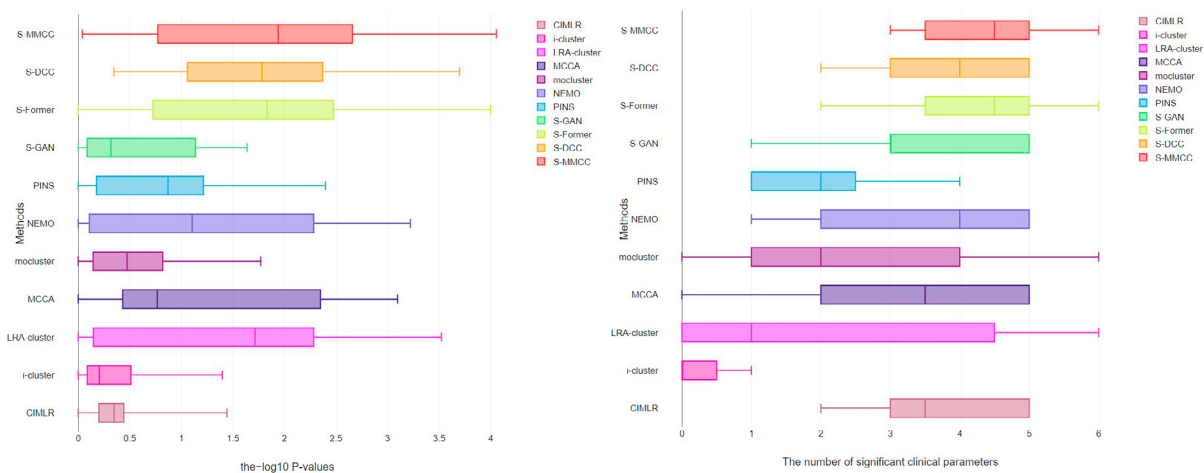


Fig. 3. Results: Survival analysis (Left) & Clinical parameters enrichment (Right)

This notable performance highlights Subtype-MMCC's effectiveness in generating distinct and cohesive clusters. Despite varying sample sizes across these cancer datasets, which significantly impact the clustering outcomes of all methods, Subtype-MMCC maintains a stable performance. It emerges as the leading method in the overall comparison of the two metrics, solidifying its position as a formidable tool in the precise subtyping of cancers.

The results, as demonstrated by the box and whisker plots in Fig 3, underscore the proficiency of Subtype-MMCC in both survival analysis and clinical parameter enrichment, which are pivotal for the accurate identification and treatment of cancer subtypes (Right plot: The -log10 P-values, Left plot: the number of significant clinical parameters). Subtype-MMCC emerges as the top-performing method, showcasing the highest median -log10 P-values in survival analysis, indicative of its superior predictive strength over competing methods. It is significant to highlight that, aside from two datasets (HNSC and PRAD), Subtype-MMCC achieved p-values close to or below 0.05 across all other datasets evaluated. This threshold is widely regarded as a benchmark for statistical significance in scientific research. Additionally, in clinical parameter enrichment, Subtype-MMCC and Subtype-Former are in a statistical tie for first place, further validating the efficacy of Subtype-MMCC in synthesizing clinical insights. Notably, it has successfully

enriched five parameters in BRCA, PRAD, and STAD datasets and all six parameters in KIRC dataset. In overall, with a comprehensive evaluation against ten other clustering methods across eight cancer datasets, Subtype-MMCC has demonstrated substantial results across all four evaluation metrics. This performance underscores the method's robustness and significant clustering capabilities.

3.2. Analysis of the contribution of different omics data to the clustering results of Subtype-MMCC

To analyze the significance of each modality derived from omics datasets, we conducted an analysis of the loss variation during training and visualized the cluster formation for each omic dataset-based modality via t-SNE graphs. This provided the opportunity to assess the prominence of each omics data type in a given cancer as well as observe its consistency/variation across different cancer types. Overall, by analyzing the loss from the 0th to the 600th epoch, it was evident that while the losses for all four modalities were similar at the start of training, towards the end, the RNA modality exhibited the lowest loss. The RNA modality loss was reduced by 16-20%, thus marking it as the most significant contributor to the variation of loss during the training process. The miRNA loss did not significantly improve during training, thus lowering its overall contribution to reducing the total ICH loss. CN and Meth modalities had similar performances with around 11-14% reduction. This pattern was fairly consistent across cancer types.

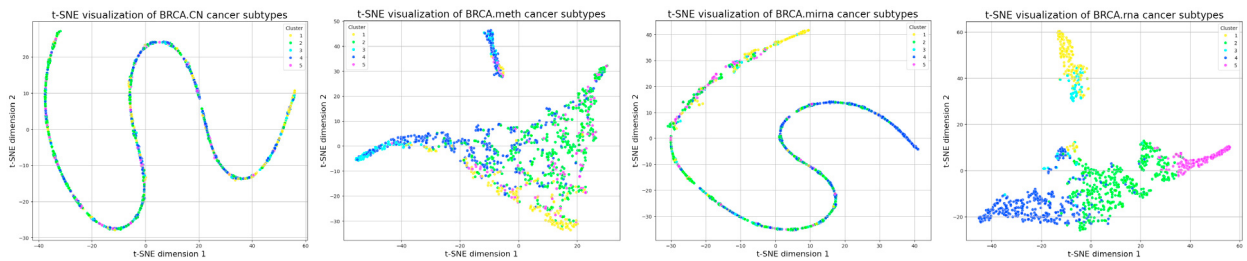


Fig. 4. t-sne graphs of CN, meth, miRNA & RNA modalities of BRCA

The t-SNE graphs plotted for each modality demonstrated the visualization of its clustering capabilities (Eg: Fig 4 : t-sne graphs of modalities of BRCA). Across all cancer types, the RNA modality consistently displayed distinctive, patterned data point arrangements, as seen in the loss analysis, although not every cancer type exhibited tight groupings or clear boundaries. For the meth modality, data points were more diffusely distributed, showing less pronounced cluster separation. CN and miRNA modalities often presented data points along a line or curve, lacking clear cluster distinction. In certain cancers, CN data did not form distinct groups, yet a gradual transition between subtypes was noticeable.

These insights are crucial for interpretive deep learning as they provide a visual understanding of the data's structure, which can be used to guide further analysis, feature selection, and model refinement. It also underscores the importance of using multimodal data for comprehensive cancer subtype characterization, as different modalities can capture different aspects of tumor biology.

3.3. Survival Analysis and Visualization of clustering results

The survival curves represent Kaplan-Meier plots, showing the probability of survival over time for different clusters representing molecular subtypes of the disease. Each line represents a cluster, with the x-axis indicating the number of days and the y-axis representing the survival probability. The survival curves generated by Subtype-MMCC provide a measurable indication of its efficacy, where the distinct survival probabilities associated with different clusters underscore its success in discerning clinically relevant subtypes (Eg: Figure 5 : Survival Analysis Curves of BRCA, HNSC, KIRC & SKCM datasets). The clearly defined and largely non-overlapping survival curves for each cluster in the datasets suggest that the model effectively differentiates between patient subgroups. The fact that different subtypes exhibit varied survival probabilities reinforces the model's effectiveness in identifying biologically and clinically meaningful categories, which could be instrumental in guiding treatment decisions and potentially improving patient outcomes.

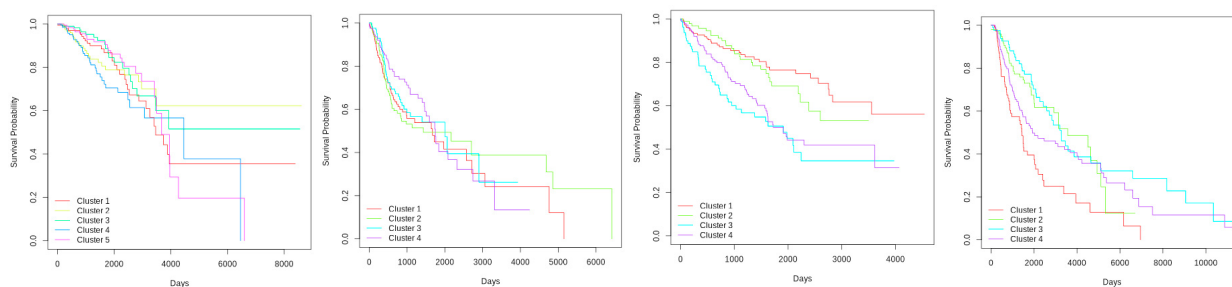


Fig. 5. Survival Analysis Curves of BRCA, HNSC, KIRC & SKCM datasets

The t-SNE visualizations of the final features post-Subtype-MMCC processing exhibit well-defined clusters that encapsulate the complex interplay of multi-omics data. As illustrated in Fig 6 for BRCA, KIRC, HNSC and LUAD cancers, Subtype-MMCC effectively delineates discrete clusters with fairly defined boundaries. Notably, the final t-SNE graphs display more pronounced clustering compared to those generated from individual omic data-type modalities (Eg: Fig 4 for BRCA). This enhanced clustering capability exemplifies the advantages and potency of leveraging multi-omics data, reinforcing the notion that a comprehensive, integrative approach to data analysis is superior to single-omics methods in capturing the true complexity of cancer biology.

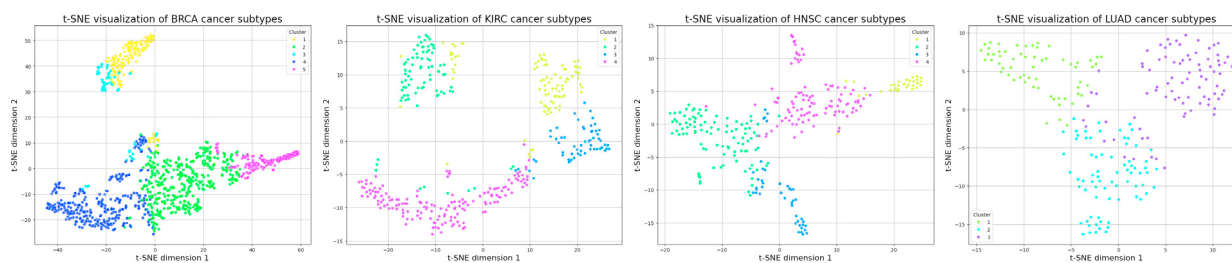


Fig. 6. t-SNE visualizations of the final features post-Subtype-MMCC processing of BRCA, KIRC, HNSC & LUAD datasets

4. Conclusion and Future work

In conclusion, Subtype-MMCC marks a significant advancement in cancer subtype identification, showcasing its robustness through an intricate multi-omics approach. By utilizing multi-modal architecture with decoupled contrastive clustering, we have developed a model with significant clustering abilities. Subtype-MMCC preserves the collective integrity of the datasets while simultaneously safeguarding the shared information across them. This ensures that the integrated characteristics are effectively encapsulated in the final clustering, which is essential for comprehensive data analysis. The efficacy of Subtype-MMCC is evident from the four evaluation metrics used, as well as from the distinct survival curves and t-SNE visualizations. In addition, Subtype-MMCC supports interpretative deep learning by revealing specific patterns and contributions from each omic dataset. This analysis helps determine the influence of each dataset on the final clustering outcome.

For future work, we plan to extend the Subtype-MMCC to identify biomarkers for each cancer subtype, analyzing them against real biomedical data. This extension can enhance cancer diagnosis and treatment, paving the way for more personalized and effective medical interventions.

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